

ETL Process Documentation

Stage 07: Indicator Analysis

1. Purpose

This document describes the Extract, Transform, Validate and Load (ETL) process used to convert the clean simulated KoboToolbox baseline dataset into analysis-ready MEAL indicators.

The process was designed to preserve data lineage and reduce manual manipulation of indicator values.

The workflow supports the project's MEAL DPro-aligned monitoring and analysis process.

2. ETL Overview

The baseline analysis follows this structure:

KoboToolbox baseline dataset ↓ Staging_Baseline_Kobo ↓ Baseline_Analysis ↓ Indicator calculations ↓ Validation ↓ Indicator summary ↓ IPTT baseline values

The Power Query transformations are performed from the baseline analysis layer rather than manually editing the source dataset.

3. Extract

Source

The source dataset was produced from the project's KoboToolbox MPCA flood-response baseline questionnaire.

The clean baseline dataset contains:

- 60 fictional households
- 165 original Kobo variables
- Household identification and demographic variables
- FCS variables
- rCSI variables
- ECMEN expenditure/resource variables
- Market-context variables
- Eligibility and assistance variables

The technical test submission was excluded from the clean analytical baseline.

Source structure

The raw Kobo data were first loaded into Excel Power Query.

The original source query was named:

Staging_Baseline_Kobo

A reference query was then created:

Baseline_Analysis

Using a Reference query preserves the connection to the staging data while allowing analytical transformations without modifying the source layer.

4. Transform

4.1 FCS Calculation

The Food Consumption Score was calculated using the standard food-group weights.

The following variables were used:

- Cereals $\times 2$
- Pulses $\times 3$
- Dairy $\times 4$
- Meat/fish/eggs $\times 4$
- Vegetables $\times 1$
- Fruits $\times 1$
- Oils/fats $\times 0.5$
- Sugar $\times 0.5$
- Condiments $\times 0$

The Power Query calculation created:

FCS_Score_PQ

The result was stored as a Decimal Number because the 0.5 weight can produce half-point scores.

A second transformation created:

FCS_Classification_PQ

Using the standard classification thresholds:

- Poor: ≤ 21
- Borderline: > 21 and ≤ 35
- Acceptable: > 35

Baseline validation produced:

- Acceptable: 52 households
 - Borderline: 7 households
 - Poor: 1 household
 - Valid households: 60
 - Acceptable FCS: **86.67%**
-

4.2 rCSI Calculation

The Reduced Coping Strategies Index was calculated using the five standard food-related coping strategies.

Weights applied:

- Less preferred food: $\times 1$
- Borrowed food / relied on help: $\times 2$
- Reduced number of meals: $\times 1$
- Reduced portion size: $\times 1$
- Adults reduced food so children could eat: $\times 3$

The Power Query calculation created:

rCSI_Score_PQ

The resulting score was stored as a Whole Number because all component weights are integers.

The existing scalar summary query:

rCSI_Baseline_Summary

calculates the mean rCSI.

Baseline result:

Mean rCSI = 10.47

4.3 ECMEN Calculation

The ECMEN calculation uses the underlying expenditure/resource variables rather than relying on a pre-calculated summary field.

This distinction was important because the Kobo raw field `ecmen_food_7d` was not populated in the clean baseline dataset.

Food expenditure

The relevant HHExpF . . . variables were dynamically selected and summed.

Created field:

ECMEN_Food_7D_PQ

Monthlyization:

ECMEN_Food_Monthly_PQ

Formula:

7-day food expenditure $\times 30 / 7$

Short-term non-food expenditure

The relevant variables beginning with HHExpNF and ending with _MN_1M were dynamically selected and summed.

Created field:

ECMEN_ShortTerm_NonFood_Monthly_PQ

These values represent the relevant 30-day expenditure period.

Long-term non-food expenditure

The relevant variables beginning with HHExpNF and ending with _MN_6M were dynamically selected and summed.

Created field:

ECMEN_LongTerm_NonFood_6M_PQ

Monthlyization:

6-month expenditure / 6

Created field:

ECMEN_LongTerm_NonFood_Monthly_PQ

Monthly household economic capacity

The three monthlyized components were combined:

ECMEN_Food_Monthly_PQ

-

ECMEN_ShortTerm_NonFood_Monthly_PQ

-

ECMEN_LongTerm_NonFood_Monthly_PQ

Result:

ECMEN_Monthly_HH_Capacity_PQ

Monthly per-capita economic capacity

Household monthly economic capacity was divided by household size.

Created field:

ECMEN_Monthly_PerCapita_PQ

Baseline mean:

58,731.20 FCFA per person per month

5. Validation

The transformed results were independently checked against the locked baseline dataset.

Validation included:

Household-level checks

- 60 valid baseline households
- Unique household IDs
- No duplicate baseline household IDs
- Required baseline eligibility fields completed
- Technical test record excluded

FCS checks

- Independent recalculation of FCS
- Classification verification
- Food-frequency range checks
- Source-field consistency
- 60/60 household calculations matched

rCSI checks

- Independent recalculation
- Standard weights verified
- Score range verified
- 60/60 household calculations matched

ECMEN checks

- Independent 7-day food expenditure calculation
- Independent monthlyization
- Independent short-term non-food calculation
- Independent long-term non-food monthlyization
- Independent household economic-capacity calculation
- Independent per-capita calculation
- 60/60 household calculations matched

The Power Query outputs therefore reproduce the validated baseline calculations.

6. Load

The final Power Query results were loaded back into Excel.

The main analysis workbook is:

Baseline_Indicator_Analysis.xlsx

The consolidated baseline summary query is:

Baseline_Indicator_Summary

It contains:

- Indicator
- Baseline value
- Unit
- Indicator direction
- Status

The summary currently contains four indicators:

Indicator	Baseline	Status
FCS acceptable	86.67%	Available
Mean rCSI	10.47	Available
Mean monthly per-capita economic capacity	58,731.20 FCFA	Available
Households at/above applicable MEB	TBD	Contextual threshold required

7. IPTT Integration

The validated baseline indicator results were then used to update the project's IPTT.

The Stage 07 IPTT contains:

- Baseline values where valid results are available
- Targets
- Actual values reserved for follow-up
- Variance reserved for follow-up

The FCS baseline was populated as:

86.7%

The rCSI baseline was populated as:

10.47

The ECMEN MEB-based baseline remains:

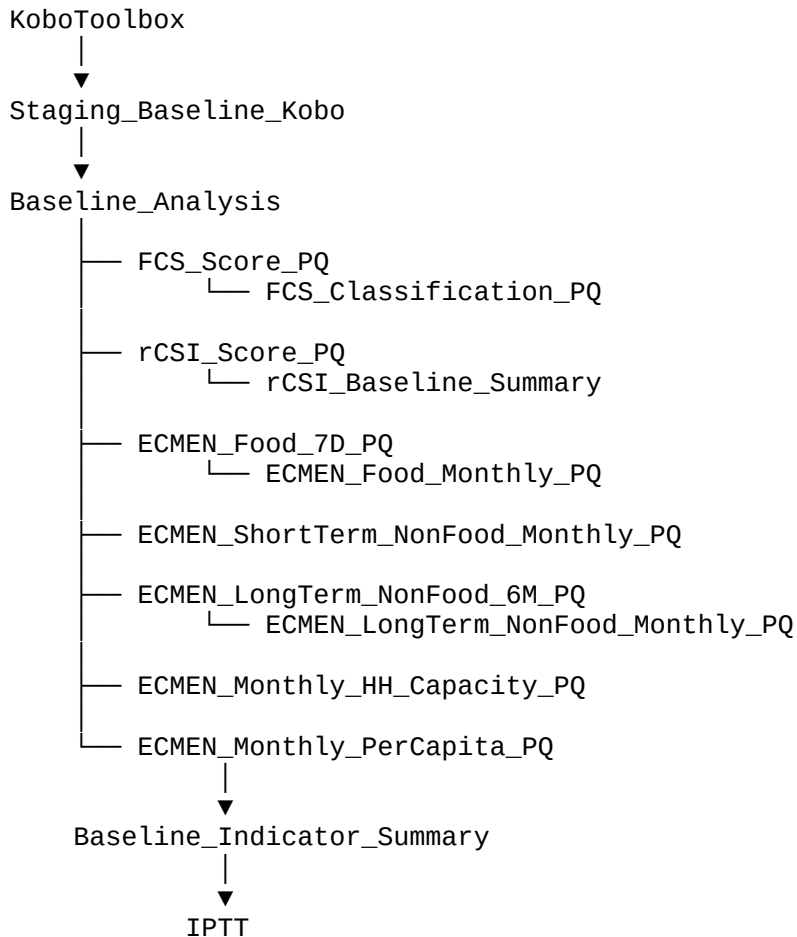
TBD

This is intentional.

No contextual MEB/SMEB has been established for this portfolio stage, so an illustrative threshold was not used to create a misleading baseline result.

8. Data Lineage

The complete baseline analysis lineage is:



9. Why Power Query Was Used

Power Query was used to make the analysis reproducible and traceable.

Instead of manually calculating and typing indicator values:

- source data remain separated from analytical transformations
- calculations can be refreshed when data change
- indicator calculations are visible in the query steps
- dynamic column selection reduces manual selection errors
- validation can be performed against the original clean dataset
- summary values remain linked to their calculation layers

This creates a clearer audit trail from raw data to programme-level indicator reporting.

10. MEAL Interpretation

The baseline values describe the situation of the simulated households at baseline.

They do not demonstrate programme impact.

A baseline establishes the starting point against which future observations can be compared.

The follow-up stage will therefore:

1. Collect or prepare follow-up observations using consistent methodologies.
2. Calculate follow-up indicator values.
3. Populate IPTT Actual values.
4. Calculate indicator-specific variance.
5. Interpret changes cautiously.
6. Identify findings requiring programme attention.
7. Document resulting decisions or adaptations.

Association between baseline and follow-up changes should not automatically be interpreted as causal programme impact.

11. Simulation and Limitations

This is a simulated portfolio project.

The 60 household records are fictional and are not statistically representative of flood-affected households in Côte d'Ivoire.

The analysis is intended to demonstrate practical MEAL skills, including:

- data preparation
- indicator calculation
- data quality assurance
- Power Query transformation
- baseline analysis
- IPTT integration
- data lineage
- interpretation and decision-use preparation

The ECMEN MEB comparison remains unresolved until an appropriate contextual MEB/SMEB is established.

12. Stage 07 Output

The main Stage 07 outputs are:

Baseline_Indicator_Analysis.xlsx

Power Query-based baseline analysis workbook.

IPTT.xlsx

IPTT containing baseline values and targets, with follow-up Actual and Variance pending.

ETL_Process_Documentation.md

This document describing the baseline ETL and indicator-analysis process.

README.md

Stage-level overview of the Indicator Analysis stage.

13. Next Stage

The next stage is:

Stage 08: Follow-Up

The follow-up process will build on the same indicator definitions and analytical logic so that baseline and follow-up results remain comparable.

The IPTT will then be updated with:

Baseline → Target → Actual → Variance

before proceeding to:

Stage 09: Findings and Decisions